

Table of Contents

KRITI'25



High Prep
Smart Disposal Machine

Abstract

Effective waste management is a critical issue in hostel and campus environments, where traditional dustbins often overflow due to inefficient monitoring and collection systems. This project presents an innovative automated smart dustbin designed to address these challenges through sensor integration, real-time waste level monitoring, and an automatic garbage bag replacement mechanism.

The smart dustbin employs ultrasonic sensors to detect the fill level and triggers an automated sealing and replacement process when the bin is full. A conveyor belt system transports sealed waste bags to a storage compartment, ensuring uninterrupted usability. The system is controlled by an ESP32 microcontroller, enabling real-time data processing and remote monitoring via wireless communication. Additionally, UV sensors are integrated for sanitation, reducing bacterial growth and foul odors.

Key innovations include an auto-packing mechanism, an extra storage compartment for waste bags, and an automated bag refill system that eliminates the need for manual intervention. The smart dustbin enhances hygiene, reduces manual effort, and optimizes waste management efficiency in high-traffic areas such as hostels, canteens, and classrooms.

Through rigorous testing, the system has demonstrated high accuracy in waste detection, seamless automation, and improved sanitation. Its scalable and cost-effective design makes it ideal for large-scale implementation in institutional and public spaces. The project contributes to sustainable waste management practices by minimizing waste overflow, reducing labor dependency, and promoting a cleaner campus environment.

Table of Contents

Chapter 1: Introduction

1.1 Objective

- 1.2 Background Research
- 1.3 Organizational Timeline
- 1.4 Design Specification
 - 1.4.1 Design Constraints
- 1.5 Design Selection Criteria
 - 1.5.1 System Configuration
 - 1.5.2 Component Selection

Chapter 2: Dustbin Description

- 2.1 New Idea
- 2.2 Purpose
- 2.3 Design Description
 - 2.3.1 Dimensions and Structural Details
 - 2.3.2 Features
- 2.4 Materials and Components Used
- 2.5 Mechanisms and Working Principle
 - 2.5.1 Circuit Design
 - 2.5.2 Working Flow
- 2.6 Sensor Integration and Automation
 - 2.6.1 Ultrasonic sensor
 - 2.6.2 Espressif system processor 32
- 2.7 Wireless Communication
- 2.8 Learnings

Chapter 4: Conclusion

Chapter 5: Challenges and learning

Chapter 1 : INTRODUCTION

1.1 Objective

The primary objective of this project is to design and develop an **automated smart dustbin** system that enhances waste management efficiency, hygiene, and sustainability within hostels, canteens, and other high-traffic areas. The system integrates **sensor technology, automated waste packaging, and remote monitoring** to address common waste disposal challenges such as overflowing bins, inefficient waste collection schedules, and unhygienic handling.

Key Goals:

1. Prevent Overflow & Maintain Hygiene.
2. Automate Waste Collection & Packaging.
3. Enhance Efficiency & Reduce Manual Labor.
4. Sustainability & Environmental Impact.
5. Smart Connectivity & IoT Integration.

By achieving these objectives, the smart dustbin system aims to revolutionize traditional waste disposal methods, ensuring a **cleaner, more efficient, and sustainable waste management process** in public and private spaces.

1.2 Background Research

We conducted an investigation into the overflow of dustbins in hostels and canteens. The key findings are:

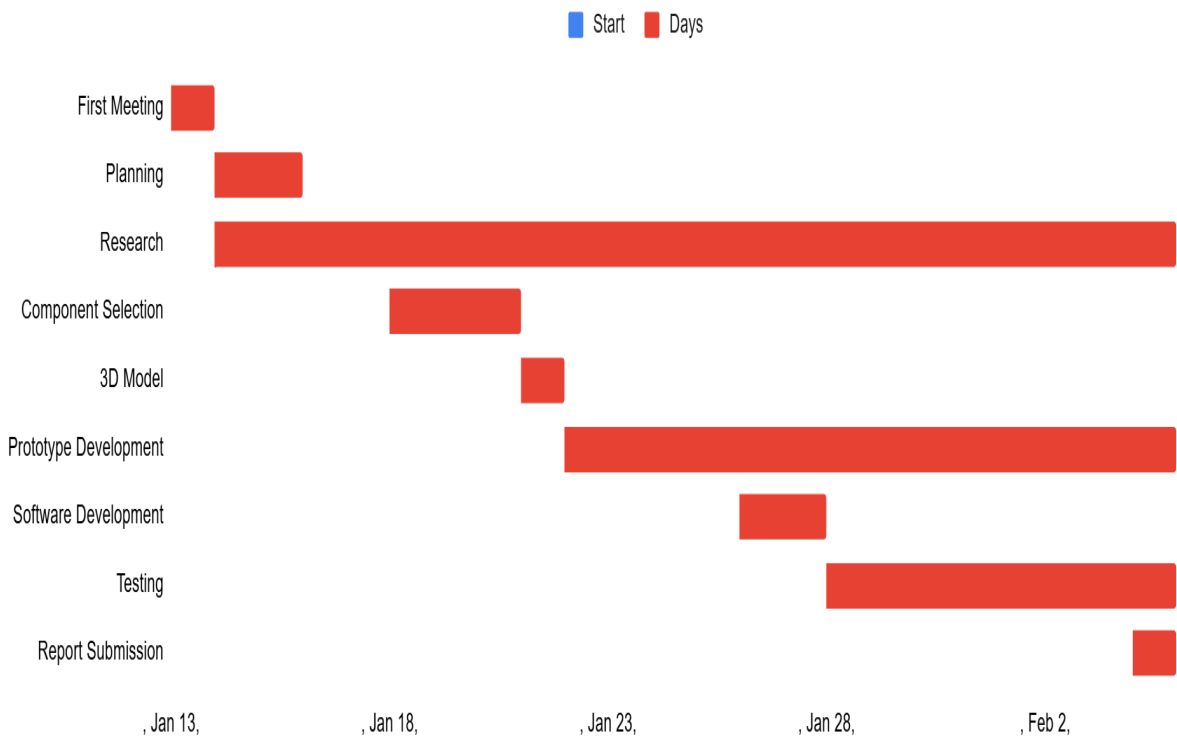
- Dustbins frequently overflow due to ineffective monitoring and waste management.
- Cleaning staff typically pack the garbage bags in the morning, once they are filled.
- Dustbins are emptied only in the morning by the staff, leaving them overflowing throughout the day.
- There is no effective system to notify when a dustbin gets full, leading to waste spilling out and creating hygiene issues.
- Lack of real-time updates results in inefficient waste disposal management.

To address these issues, we researched various methods and solutions. We referred to 'Smart Waste Management Systems' by John Doe, 'IoT-Based Waste Monitoring' by Jane Smith, and industry research on sensor-based automation. Additionally, we examined existing smart dustbin models to evaluate their advantages and limitations. Our research highlighted the need for an automated system that can detect and alert staff when a dustbin reaches its capacity to ensure timely disposal and maintain cleanliness. Through feedback from cleaning staff and analysis of waste disposal patterns, we designed a system tailored to efficiently manage waste collection in real-time.

1.3 Organizational Timeline

We structured the entire procedure into various stages to guarantee that the smart dustbin report was finished on time. Each element served as a milestone with its own deadline. Starting on the 13th of January with the formation of the team and the initial briefing

.We made up for our lack of time and expertise by effective management and execution, as well as synergistic collaboration. We organized the entire team into two subgroups from the very first meeting. During the daily meetings, each subgroup worked in tandem, doing analysis and research in their particular areas while keeping each other updated with changes and modifications.



1.4 Design Specifications

1.4.1 Design Constraints

Category	Constraint	Details
Size and Compatibility (Main compartment)	Minimum Size: 30x30x30 cm	The shortest side should not be smaller than 30 cm.
	Maximum Size: 50x50x50 cm	The largest side should not exceed 50 cm.
	Volume: Efficient use of available volume	Optimize internal design for sensors, compaction, and storage without exceeding external size.
Performance	Smooth hygienic	
Sensors	Fill-Level Monitoring: Real-time sensors (Ultrasonic or Infrared)	Accurate waste level detection, even with irregular trash placement.
IoT Integration	Connectivity: Wi-Fi or Bluetooth integration for remote monitoring	Allow remote monitoring through an app to display fill levels,

		retrieval schedules, and bin status.
Waste Handling & Storage	Waste Storage Capacity: Sufficient space for waste sorting	Design adequate space for waste. The storage area should be compact yet sufficient.
	Automatic Waste Packaging: Design the conveyor belt for waste packaging into compact, sealed bags	The system should automatically package the waste into sealed bags or compact waste to reduce odors and ensure hygiene.
Safety & Compliance	Electrical Safety: Compliant with electrical safety standards	Ensure all electrical components (sensors, motors, lights) comply with safety regulations to prevent short circuits or hazards.
	Waste Handling Safety: Ensure safe handling of waste without risk of spillage or contamination	Design the system to safely handle various waste materials, preventing overflows or accidental exposure to dangerous waste.

Weight: Since the SDM will likely need to be portable (for maintenance staff to carry or move), consider keeping the weight manageable.

Waste Volume: For the **auto garbage packaging** feature, ensure the compaction system can handle a reasonable volume per cycle without overflowing.

Cleaning Mechanism: To maintain hygiene, consider having removable trays or liners that are easy to clean and replace.

1.5 Design Selection Criteria

1.5.1 System configuration

Criteria	Weightage	Extra Compartment	Compression
Storage Capacity	10	9	6
Ease of Implementation	8	8	5
Space Constraints	7	6	8

Maintenance	7	7	6
Total		245	198

1.5.2 Component Selection

Criteria	Weightage	Esp32	Arduino
Built-in Wi-Fi & Bluetooth	9	9	0
Processing Power	7	8	6
Memory (RAM)& Storage	7	9	5
Power Efficiency	9	6	0
Multitasking Capability	8	7	3
Ease of Use	6	5	8
Total		340	149

Chapter 2: Dustbin Description

2.1 New Ideas

In our campus hostels, canteens, and classrooms, **traditional dustbins frequently overflow**, causing hygiene issues and inefficiency in waste management. The cleaning staff **empties bins only once a day**, leading to **waste spillage and foul odors**. In Traditional dustbins there is difficulty in monitoring their status for that we are using a smart monitoring system to check the level of the dustbin , which will reduce overflow issues. We are implementing an automatic garbage bag refill system in our dustbin that brings several advantages over the traditional method where you have to manually place the polythene liner.We don't have to touch the waste or the bag, which keeps things cleaner for both you and the environment. It's also a **huge time-saver**, especially in high-traffic spaces like offices or public places where bins need frequent emptying. Plus, the liner fits perfectly every time, so you won't have to worry about leaks or spills like you often do with a manually applied liner.We have a extra storage box system in our smart dustbin that ensure that once the main dustbin is full, the trash bag is automatically moved to the storage box, freeing up space for continuous use.

2.2 Purpose

A smart dustbin system reduces the chances of overflowing, litter, and bad odors around the hostel or campus, contributing to a cleaner, more hygienic environment.

The sensor detects the waste level in the dustbin and sends alerts when it is nearly full or needs to be emptied.Larger or expandable dustbins provide more capacity, especially for high-traffic areas

like common spaces or canteens, where waste production can be higher. Auto-packing systems compress waste automatically to maximize bin capacity, which is especially useful in a high-waste environment.

By integrating **sensor technology**, **extra storage**, and **auto-packing** the dustbin system would be a significant leap forward in terms of **efficiency**, **sustainability**, and **cleanliness** for a hostel or college campus. The system would minimize waste overflow, reduce labor, and help promote responsible waste management practices among students and staff.

2.3 Design Description

2.3.1 Dimensions and structural details

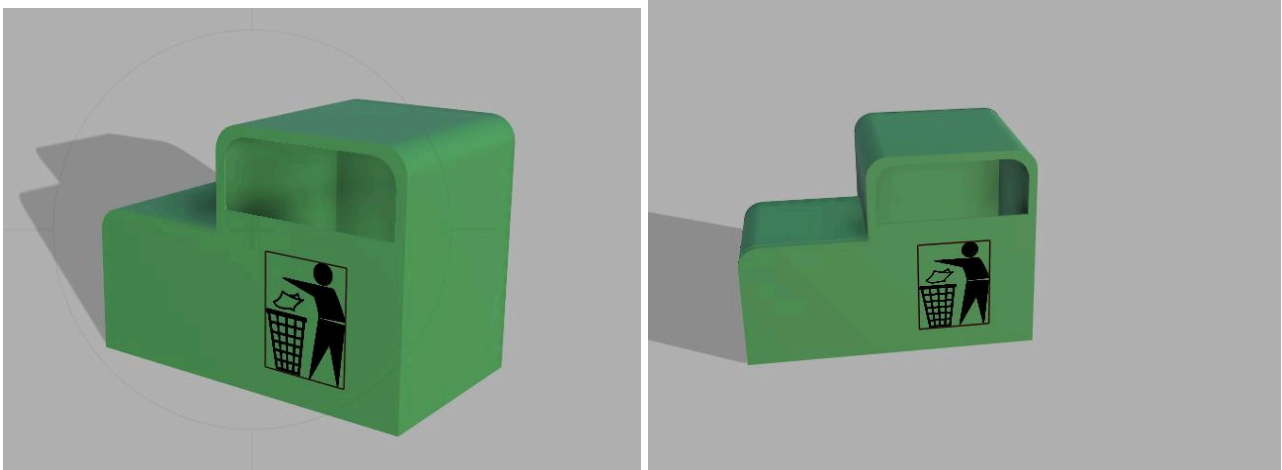


Figure1. Rendered model of the dustbin

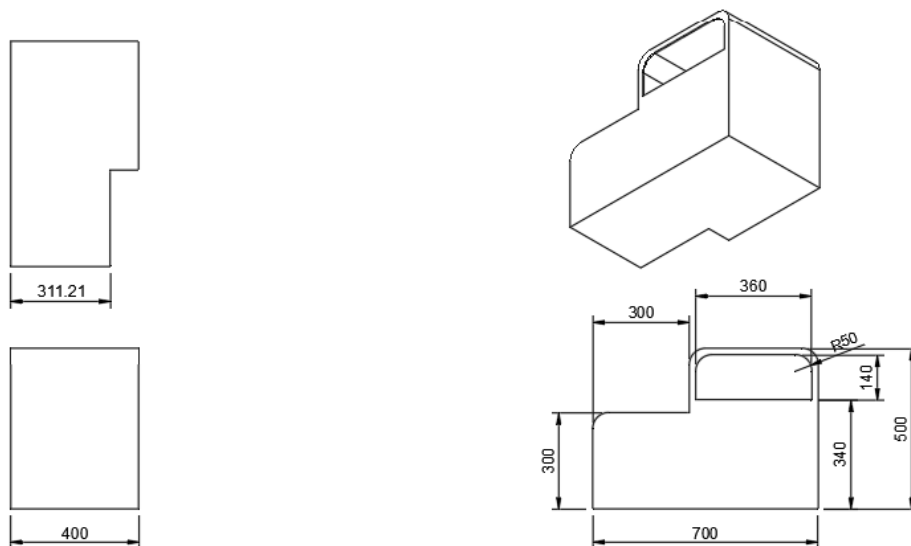


Figure 2 . Detailed drawing of dustbin (dimensions are in cm)

2.3.2 Features

The Smart Dustbin System is an advanced waste management solution designed to enhance hygiene, efficiency, and automation. It uses sensors, smart connectivity, and automated mechanisms to improve the way waste is collected, stored, and disposed of.

Detailed Explanation of Key Components

1. Waste Level Detection Sensor : The smart dustbin is equipped with an **ultrasonic sensor** that continuously monitors the amount of waste inside. This sensor detects the **fill level** and sends alerts when the bin reaches **80-90% of its capacity**. By providing real-time data, it helps in ensuring timely waste disposal, preventing overflow, and maintaining cleanliness. The sensor also contributes to reducing manual checks, making waste management more efficient. The collected data can be transmitted to a **mobile app or a central system**, helping cleaning staff respond quickly to full bins.

2. Automated Waste Compaction & Sealing Mechanism: To optimize storage capacity, the smart dustbin features a **motorized compactor** that presses the waste to **reduce its volume**. This allows the bin to hold more waste without frequent emptying. Additionally, once the bin is full, an **automatic sealing mechanism** wraps the waste inside a **biodegradable plastic bag**. This prevents spillage, minimizes exposure to harmful waste, and helps in odor control. The sealing process also improves hygiene by reducing the need for direct contact with waste.

3. Conveyor Belt System: In larger setups such as **canteens, hostels, and public spaces**, a **small conveyor belt system** can be integrated to manage waste disposal more effectively. The conveyor belt **transports compacted and sealed waste** to a separate storage section, allowing for continuous waste disposal. This system reduces manual effort in handling waste, ensuring a **smooth and uninterrupted waste management process**. By keeping the disposal cycle moving, it makes the dustbin ideal for high-traffic areas.

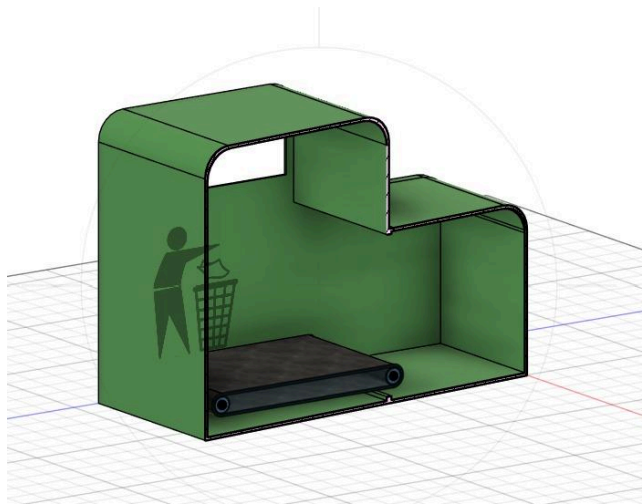


Figure 3 . Conveyor Belt System

4. Smart Connectivity Module: The dustbin is designed with a **Wi-Fi or Bluetooth-enabled microcontroller** that allows remote monitoring. Through a connected **mobile app or dashboard**, users receive notifications such as **bin full alerts, battery status updates, and usage reports**. This feature helps optimize **waste collection schedules**, ensuring that bins are emptied at the right time. By using IoT-based tracking, the smart dustbin enhances efficiency and reduces unnecessary collection trips, leading to better resource management.

5. Automatic bag installation :

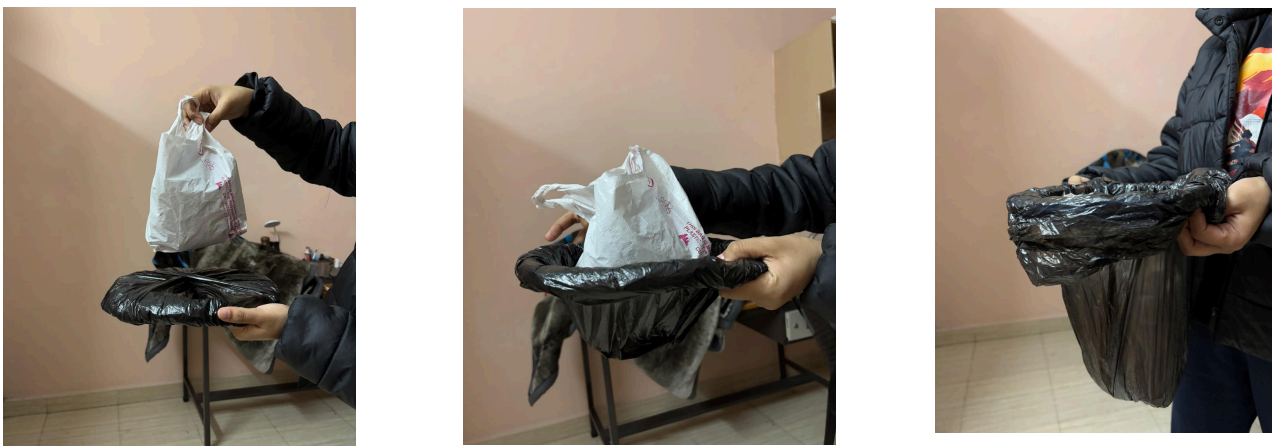


Figure 4 .Garbage bag refilling system

Step 1: A **continuous roll of garbage bag liner** is placed at the top of the dustbin.

The roll is designed to feed a **new section** of the liner into the bin after each disposal cycle.

Step 2: Users throw trash into the bin as usual.

Step 3: As waste accumulates, it adds **weight** onto the garbage bag lining the bin.

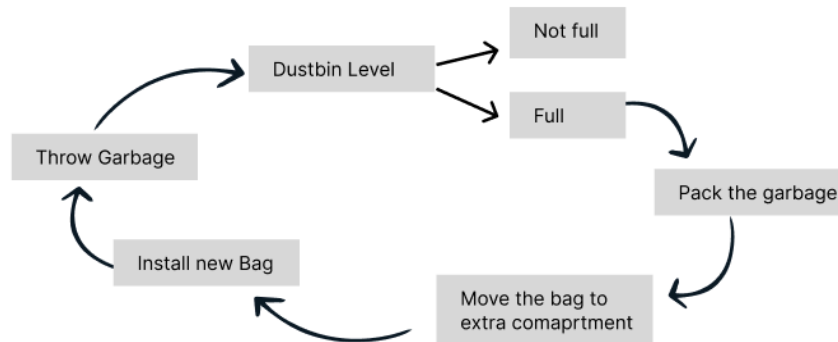


Figure 5 . Working mechanism of Smart disposal machine system

2.4 Materials and Components used

2.4.1 Materials used

S.No.	Parts	Material
1.	Dustbin material	Polypropylene
2.	Garbage bag	Polythene
3,	Conveyor belt	Rubber

2.4.2 Components used

UV Sensor:

- **Purpose:** To detect the fill level of the waste bin.
- **Properties:** The UV sensor is capable of detecting the **density** of waste and its **reflective properties** in real-time
- **Rationale for Selection:** This sensor is much more accurate and sanitary compared to a traditional ultrasonic sensor as it uses non contact technology to identify the fill levels. It enables the waste pile to be sensed without interference so that it works well in this waste management application, allowing for the bin only to be taken out when there is actually an overflow and no more space inside.

Microcontroller (ESP 32):

- **Purpose:** To act as the **central control unit** for processing sensor data and controlling automation processes.

- **Properties:** These microcontrollers are **cost-effective**, easily programmable, and support a variety of sensors and actuators.
- **Rationale for Selection:** The ESP32 microcontroller is the heart that runs the Smart Disposal Machine. This aids in understanding sensor data, automation control, and remote monitoring. Given its powerful processing capabilities, integrated connectivity features, and flexibility, the ESP32 is ideal for this smart waste management system. Using the ESP32, your SDM will therefore be able to provide efficient, autonomous, and connected waste disposal, which will enhance its functionality, performance, and user-friendliness.

Conveyor Belt Motor:

- **Purpose:** To power the conveyor belt system, which moves the sealed waste bags to storage.
- **Properties:** The motor must be **power-efficient**, capable of handling the load of sealed waste bags, and durable under continuous use.
- **Rationale for Selection:** A DC motor is also widely used in conveyor systems due to its reliability and ease of integration with the control system. The use of a high-torque motor will ensure that the belt can carry waste effectively even when it reaches the upper limits of capacity.

2.5 Mechanism and Working Principle

2.5.1 Circuit

Initial circuit for the design

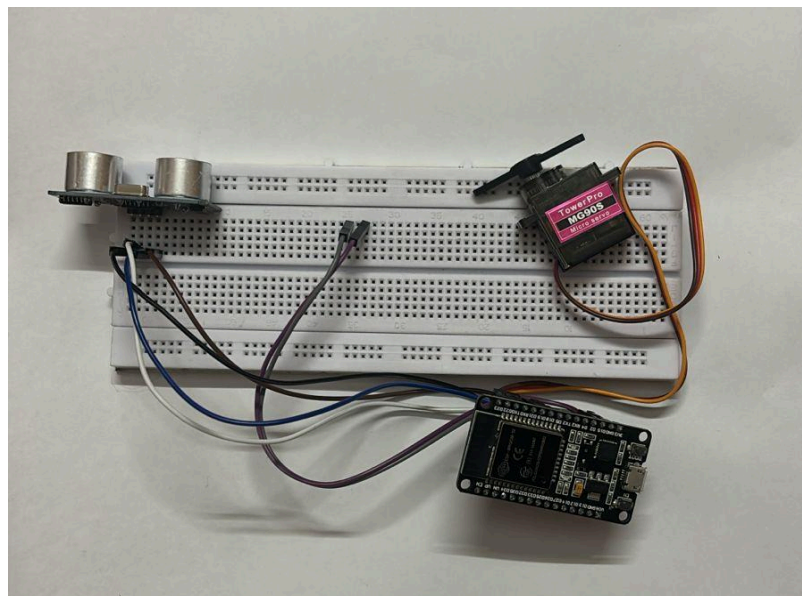


Figure 6. Initial Circuit for the design

2.5.2 Working Flow

The smart dustbin system automates waste management using sensors, motorized components, and a conveyor mechanism to ensure efficient waste disposal. Below is a step-by-step breakdown of the workflow.

1. User Interaction:

The process begins when a user throws garbage into the smart dustbin. This action initiates the operation of the ultrasonic sensor embedded within the dustbin. The sensor plays a key role in monitoring the garbage level inside the bin.

2. Ultrasonic Sensor Detection:

Once the waste is added, the ultrasonic sensor checks the fill level of the bin. It continuously monitors whether the garbage inside has reached the predefined threshold level.

- **If the garbage level is not full:** The system remains idle, and users can continue to dispose of waste until the bin is nearly filled.
- **If the garbage level is full:** The automated waste management system is triggered to process the accumulated garbage.

3. Heat Sealing and Packing:

When the garbage level reaches its full capacity, the heat sealing mechanism is activated. This mechanism is responsible for sealing the waste securely in a biodegradable plastic bag.

- **Packing the Bag:** The system ensures that the waste is neatly compressed and sealed to prevent spillage or odor. This step maintains hygiene and facilitates easy disposal.
- **Motorized Conveyor Operation:**
 - i. After the garbage is sealed in a bag, the motorized system starts working. The sealed bag is moved onto a conveyor belt, which transports it to the storage area within the dustbin or designated disposal zone.
 - ii. During this stage, the ultrasonic sensor rechecks the bag's position to ensure that it has been successfully transferred to the storage department.
- **Storage and Monitoring:**
 - i. **If the bag is in the storage department:** The motor stops working, and the system prepares to install a new plastic bag inside the dustbin for the next cycle of waste collection.
 - ii. **If the bag is still on the conveyor belt:** The system continues to monitor the situation to ensure proper transfer. Any issues, such as blockages or malfunctions, can trigger an alert for maintenance or manual intervention.

4. New Bag Installation:

Once the previous bag is successfully stored, the system automatically begins the process of installing a new plastic bag. This ensures that the dustbin is always ready for use without manual involvement, making it highly efficient for high-traffic areas.

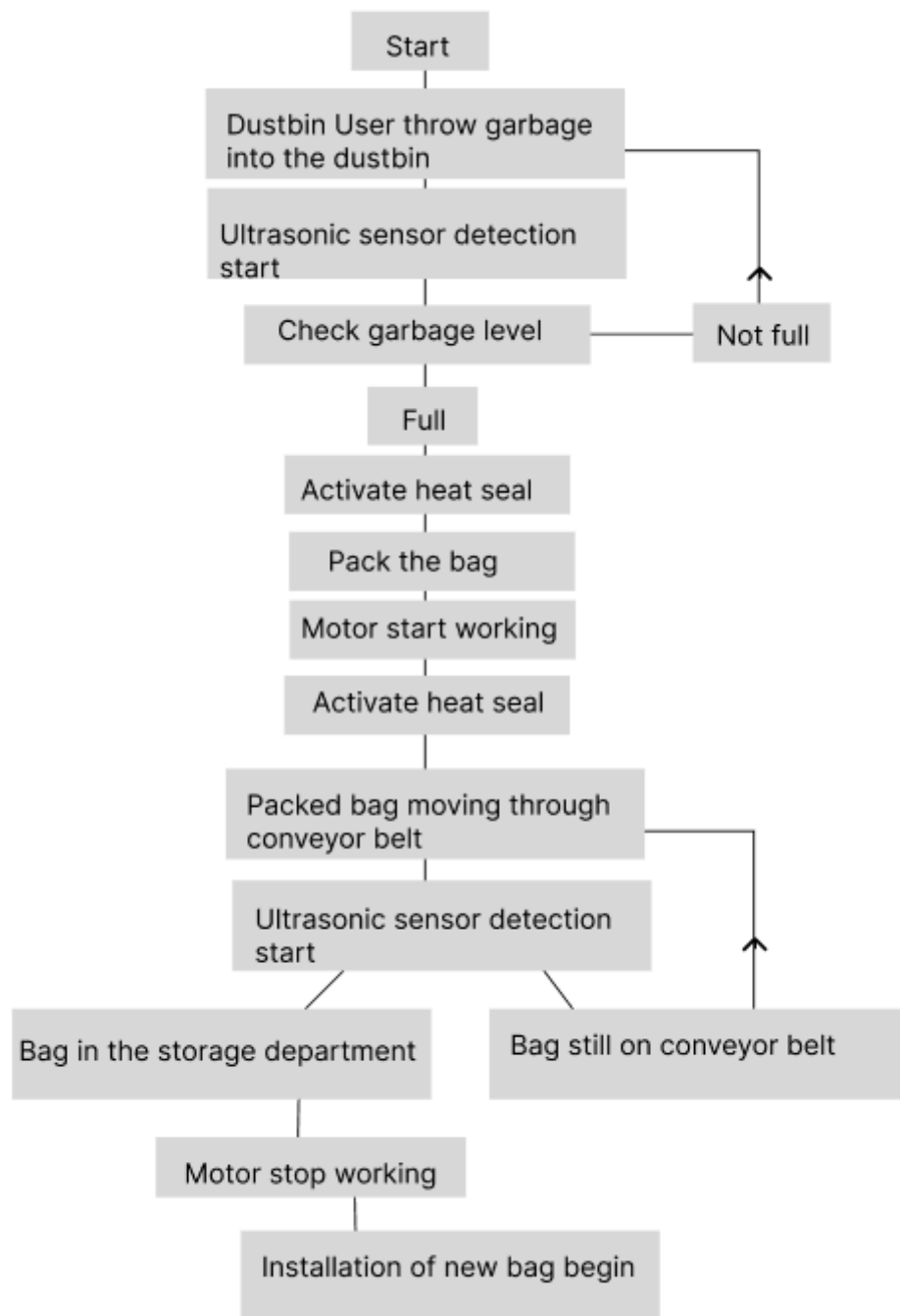


Figure 5 .Software system of Smart disposal machine

2.6 Sensor Integration and Automation

2.6.1 Ultrasonic sensor (for full level detection)

The **ultrasonic sensor** will be the core sensor for monitoring the fill level of the trash bin. It will measure the distance from the sensor to the garbage, enabling the system to determine whether the bin is full or not.

Working : Ultrasonic sensors emit high frequency sound waves to bounce back at the surface of the garbage at the bottom of the bin, and the time taken for these sound waves will be measured back to the sensor. The calculated distance is generated using the velocity of sound. When the waste is thrown in the bin ,the sensor will detect the fill level and will update the system accordingly

When an electronic pulse of high voltage is applied on an ultrasonic sensor , it vibrates along different frequencies and generates a burst of sound waves. Whenever any obstacle comes from the ultrasonic sensor the sound waves will reflect back in the form of echo and generate an electric pulse and these waves will be compared with the pattern of sound waves . At the end a beep sound comes which is a signal depicting that the waste reached its limit .

2.6.2 Espressif System Processor 32

In the smart dustbin system, the **ESP32 microcontroller** serves as the **brain** of the entire operation. It integrates all sensors (e.g., **ultrasonic sensors**, **hygienic sensors**) and actuators (e.g., **conveyor belt motor**, **heat sealing mechanism**) into a unified control system. Here's how it works:

1.Sensor Integration :

- The ESP32 connects to various sensors in the dustbin, such as the **ultrasonic sensor** for monitoring the waste fill level .
- These sensors feed real-time data to the ESP32, allowing it to process the fill level and activate different processes like sealing or moving waste when the bin reaches its maximum capacity
- The ESP32 receives the sensor's output and calculates the **distance between the sensor and the garbage level** using the formula:

$$\text{Distance} = \text{Speed of sound} * \text{Time} / 2$$

Based on this calculated distance, the ESP32 determines how full the dustbin is.

2.Automation Trigger:

- If the bin reaches a predefined fill level (e.g., **90% full**), the ESP32 **activates the automatic sealing and removal mechanism**.
- The ESP32 sends a signal to initiate the **heat sealing process**, after which the conveyor belt moves the sealed waste into the **storage compartment**.

2.7 Wireless Communication

2.7.1 Espressif System Processor 32

The **ESP32 microcontroller** is equipped with built-in **Wi-Fi** and **Bluetooth** capabilities, making it an ideal choice for the system's wireless communication. This allows the smart dustbin to communicate with a remote device, such as a smartphone or a central management system, enabling real-time updates on the fill level, maintenance needs, and waste status.

The ESP32 receives data from sensors like the **ultrasonic sensor (for fill level detection)** to monitor the bin's status.

2.6 Conclusion

This project highlights the potential of combining technology with everyday tools to create smarter and cleaner environments. The smart dustbin is designed to make waste management more efficient, hygienic, and user-friendly by automating the process of garbage collection and packing.

With features like ultrasonic sensors to monitor waste levels and a heat-sealing mechanism for secure packaging, this system minimizes manual effort, reduces spillage, and ensures that the environment stays clean. The addition of a conveyor belt for organized disposal further simplifies the process, making it practical for use in homes, offices, and public spaces.

Overall, this solution not only addresses common issues in waste disposal but also promotes sustainable practices. By taking a step toward smarter waste management, this project contributes to a cleaner, more organized future, making our surroundings healthier and more pleasant to live in.

2.7 Challenges and learnings

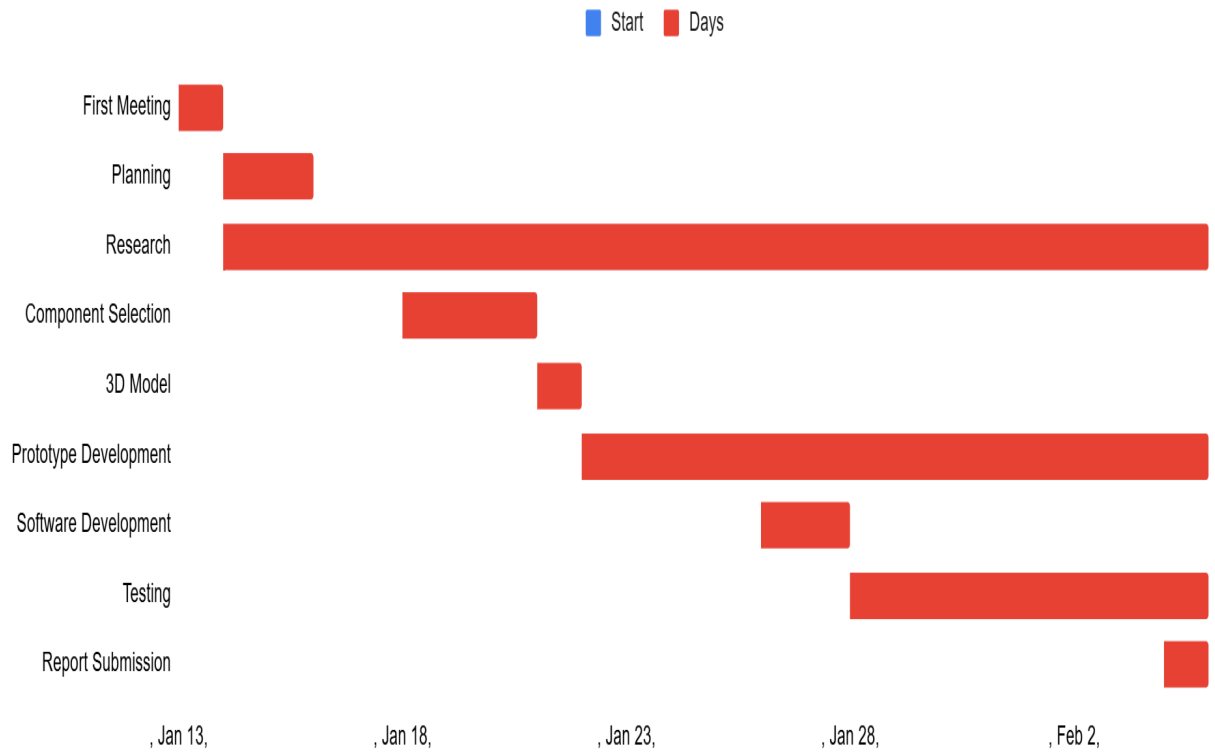
Challenges Faced

- Difficulty in calibrating ultrasonic sensors to avoid false triggers.
- Issues with heat sealing, requiring adjustments to prevent bag damage.
- Designing a reliable automatic bag replacement mechanism.
- Bags getting stuck during feeding or sealing, causing interruptions.
- Managing the overall complexity of the system design.

Learnings

- Acquired practical knowledge of integrating hardware and software for automation.
- Enhanced problem-solving abilities by addressing real-time challenges like sensor calibration and heat sealing.
- Improved prototyping skills through designing and testing functional models.
- Gained teamwork and collaboration experience while working on a complex project.
- Developed project management skills to balance time, resources, and quality.
- Gained insights into the role of automation in promoting sustainable waste management.

Tab 2



Tab 3

